Proton Transfer, Strong & Weak Acids

Exam code: 0620 & 0971

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Updated on 24 September 2024

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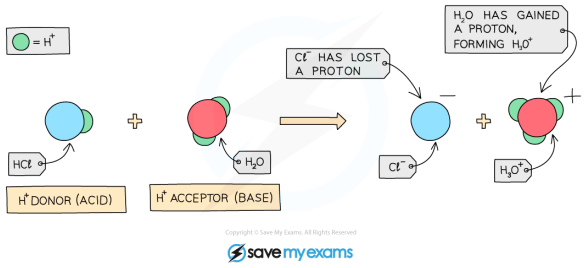
Proton transfer, strong & weak acids

Proton transfer

Extended tier only

- Acids are **proton donors** as they ionise in solution producing protons, which are H^+ ions
- These H^+ ions make the aqueous solution acidic
- Bases are **proton acceptors** as they accept the protons which are donated by the acid

Proton transfer between hydrochloric acid and water



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What is a strong acid?

- Acids can be either **strong** or **weak**, depending on how many H^+ ions they produce when dissolved in water
- Strong acids **completely dissociate** (or ionise) in water, producing solutions of a **very low pH**
- Strong acids include HCl and H_2SO_4
- Example of a strong acid: hydrochloric acid

$$HCl\ (aq) \rightarrow H^+\ (aq) + Cl^-\ (aq)$$

What is a weak acid?

- Weak acids **partially dissociate** (or ionise) in water and produce pH values which are closer to the **middle** of the pH scale, whilst still being below 7
- For weak acids, there is usually an equilibrium set-up between the molecules and their ions once they have been added to water
- Example of a weak acid: propanoic acid

$$CH_3CH_2COOH \rightleftharpoons H^+ + CH_3CH_2COO^-$$

- The equilibrium lies to the **left**, indicating a high concentration of intact acid molecules, with a low concentration of H^+ ions in the solution
- Another example of a weak acid is ethanoic acid which will react with alkalis such as sodium hydroxide to form ethanoate salts

ethanoic acid + sodium hydroxide → sodium ethanoate + water

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$$

Effect of concentration on strong and weak acids

- A concentrated solution of an acid is one that contains a **higher number** of acid **molecules** per dm^3 of solution
- It does not necessarily mean that the acid is strong though, as it may be made from a weak acid which does not **dissociate** completely
- For example a dilute solution of HCl will be more acidic than a concentrated solution of ethanoic acid, since most of the HCl molecules dissociate but very few of the CH_3COOH do

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